

EEE 103 - Computer Programming

L8 - Strings/Char Arrays

Part 2: Strings in C

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Strings in C

- A string is a character array terminated by null character '\0'
- String literals automatically include the null terminator
- You can access characters using index: str[0], str[1], ...
- Example:

```
char str[6] = "Hello";  
// Memory: H e l l o \0
```

Declaring and Initializing Strings

Different ways to create strings

```
char name[20];  
char str[] = "Hello";  
char msg[6] = "World";  
char arr[] = {'H', 'i', '!', '\0'};
```

Key Points:

-  char str[] = "text"; → size includes \0 automatically
-  String literals add \0 at the end
-  Manual array needs explicit \0
-  Always reserve space for null terminator

Taking String Input

- Never use & with string variables
- `scanf("%s", str);` → stops at space
- `gets()` → deprecated and unsafe
- `fgets()` → safe and recommended

```
scanf("%s", name); // No & needed, can't read spaces
```

```
gets(line); // Deprecated & unsafe!
```

```
fgets(line, sizeof(line), stdin); // Safe + reads spaces
```

Printing Strings

- Use %s to print full string until \0
- Print character by character with loop
- %s stops automatically at null terminator

```
printf("%s", str); // Entire string
```

```
for(int i = 0; str[i] != '\0'; i++)  
    printf("%c ", str[i]);
```

Count Total Characters

Loop until \0

```
1 #include <stdio.h>
2 int main() {
3     char str[100]; int count=0;
4     printf("Enter a string: ");
5     scanf("%s", str);
6     while(str[count] != '\0') count++;
7     printf("Total characters: %d\n", count);
8     return 0;
9 }
```

Output:

```
Enter a string: Programming
Total characters: 11
```

Reverse String

Input: ABCD → Output: DCBA

```
1 #include <stdio.h>
2 int main() {
3     char str[100]; int len=0;
4     printf("Enter a string: "); scanf("%s", str);
5     while(str[len] != '\0') len++;
6     for(int i=len-1; i>=0; i--) printf("%c", str[i]);
7     return 0;
8 }
```

Output:

Enter a string: ABCD
DCBA

Count Words

Using fgets()

```
1 #include <stdio.h>
2 int main() {
3     char str[200]; int words=1;
4     printf("Enter a string: ");
5     fgets(str, sizeof(str), stdin);
6     for(int i=0; str[i]!='\0'; i++)
7         if(str[i]==' ') words++;
8     printf("Total words: %d\n", words);
9     return 0;
10 }
```

Output:

```
Enter a string: This is a test string
Total words: 5
```

Concatenate Strings (Manual)

```
1 #include <stdio.h>
2 int main() {
3     char s1[100], s2[50]; int i=0,j=0;
4     printf("First: "); scanf("%s", s1);
5     printf("Second: "); scanf("%s", s2);
6     while(s1[i] != '\0') i++;
7     while(s2[j] != '\0') s1[i++] = s2[j++];
8     s1[i] = '\0';
9     printf("Result: %s\n", s1); return 0;
10 }
```

Output:

```
First: Hello
Second: World
Result: HelloWorld
```

Character Frequency

"This is a test" → 'i' appears 3 times

```
1 #include <stdio.h>
2 int main() {
3     char str[100], ch; int count=0;
4     printf("Enter string: ");
5     fgets(str, sizeof(str), stdin);
6     printf("Enter character: "); scanf("%c", &ch);
7     for(int i=0; str[i]!='\0'; i++)
8         if(str[i]==ch) count++;
9     printf("Frequency of '%c' = %d\n", ch, count);
10    return 0;
11 }
```

Output:

```
Enter string: This is a test string
Enter character: i
Frequency of 'i' = 3
```

Practice Problems

1

Convert string to uppercase

2

Count word occurrences

3

Check substring

4

Extract substring

5

Count vowels & consonants

6

Palindrome check

7

Remove spaces

8

Reverse words in sentence

Thank You!